



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

JUNE 2005

TASK #3

YEAR 12

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 2 Hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

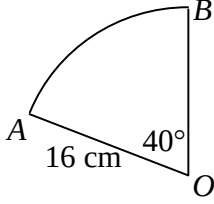
Total Marks - 100 Marks

- Attempt Questions 1 - 6
- All questions are NOT of equal value.

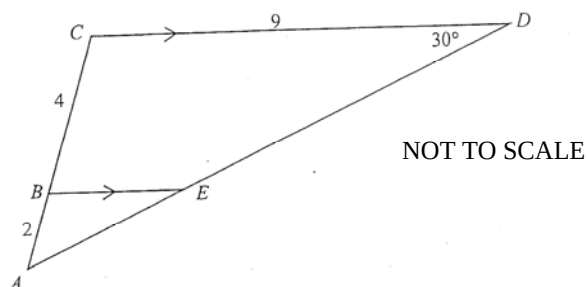
Examiner: *P. Bigelow*

Total marks – 100
Attempt Questions 1 - 6
All questions are NOT of equal value

Answer each SECTION in a SEPARATE writing booklet.

Section A		Marks
Question 1 (15 marks)		
(a)	Convert 150° to radians	1
(b)	 OAB is a sector of a circle, with radius 16 cm. Write down the area of the sector OAB, correct to the nearest square centimetre.	2
(c)	Write down the value of the following, correct to 2 decimal places.	
(i)	$\frac{1}{\sqrt{e}}$	1
(ii)	$\cos 3$	1
(d)	Write down the derivatives of the following	
(i)	e^{-4x}	1
(ii)	$\tan\left(\frac{x}{2}\right)$	1
(e)	Thirty cards are numbered from 1 to 30.	
(i)	If one card is selected at random what is the probability that the card has a 4 on it?	1
(ii)	If two cards are chosen at random, what is the probability that at least one of the cards has a 4 on it?	2
(f)	Find $f(2)$ if $f'(x) = 2x + 3$ and $f(1) = 6$	2

(g)



In the diagram, ACD is a triangle where $AB = 2$ cm, $BC = 4$ cm, $CD = 9$ cm and $\angle CDE = 30^\circ$. Also BE is parallel to CD .

- | | | |
|------|---|---|
| (i) | Find the size of $\angle BED$. | 1 |
| (ii) | Find the length of BE . Give reasons for your answer. | 2 |

Question 2 (17 marks)

- | | | |
|-------|----------------------------------|---|
| (a) | Differentiate the following | |
| (i) | $2x \ln x$ | 2 |
| (ii) | $\frac{\cos x}{x}$ | 2 |
| (b) | Find | |
| (i) | $\int_1^4 \frac{dx}{\sqrt{x^3}}$ | 2 |
| (ii) | $\int \frac{x^3 + x}{x^2} dx$ | 2 |
| (iii) | $\int (e^x + 1)^2 dx$ | 2 |

Question 2 continues over the page

Question 2 continued

- (c) A farmer made the following statement:
“In 1998 the price of wheat started to fall and it reached its lowest levels in mid 2002. Then from mid 2002, the price began a continual rise and is now, in 2005, at an all time high.”

If the curve representing the price of wheat, P , is differentiable, what does the above statement imply about:

- | | | |
|------|---------------------------------------|---|
| (i) | $\frac{dP}{dt}$ between 1998 and 2005 | 2 |
| (ii) | $\frac{d^2P}{dt^2}$ at mid 2002? | 1 |
- (d)
- | | | |
|------|--|---|
| (i) | Sketch $y = 3\sin 2x$ for $0 \leq x \leq 2\pi$ | 2 |
| (ii) | Find $\int_0^{\frac{\pi}{4}} 3\sin 2x \, dx$ | 2 |

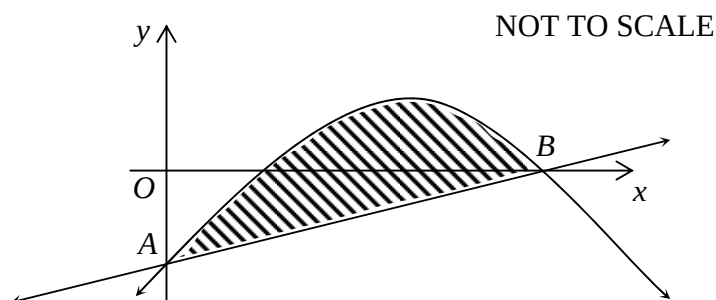
End of Section A

Section B (Use a SEPARATE writing booklet)

Question 3 (17 marks)

Marks

- (a) The graphs of $y = x - 4$ and $y = -x^2 + 5x - 4$ intersect at the points A and B , as shown in the diagram below.

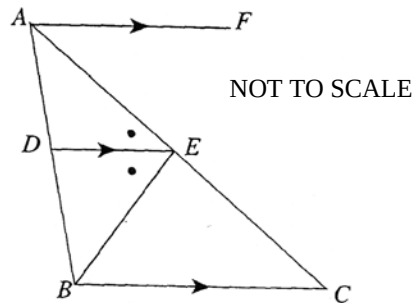


- | | | |
|-------|--|---|
| (i) | Find the coordinates of A and B | 2 |
| (ii) | Find the area of the shaded part. | 2 |
| | | |
| (b) | Find the equation of the tangent to $y = \ln x$ at the point where $y = \ln x$ crosses the x axis. | 2 |
| | | |
| (c) | Consider the curve given by $y = x^3 - 6x^2 + 9x + 6$ | |
| (i) | Find the coordinates of the stationary points and determine their nature. | 3 |
| (ii) | Find the coordinates of any points of inflexion. | 1 |
| (iii) | Sketch the curve for $-1 \leq x \leq 4$. | 2 |
| (iv) | Indicate on your curve, with the letter S , where the curve has the greatest rate of increase. | 1 |

Question 3 is continued over the page

Question 3 continued

(d)



The diagram shows $\triangle ABC$ with D lying on AB and E lying on AC .

The lines AF , DE and BC are parallel and $\angle AED = \angle BED$.

- | | | |
|-------|--|---|
| (i) | Show that $\triangle BEC$ is isosceles. | 2 |
| (ii) | State the reason why $AD : DB = AE : EC$. | 1 |
| (iii) | Show that $AD : DB = AE : EB$ | 1 |

Question 4 (16 marks)

- | | | |
|-----|--|---|
| (a) | The part of the curve $y = \frac{1}{\sqrt{2x+1}}$ between $x = 0$ and $x = 1$ is rotated about the x axis. | 3 |
|-----|--|---|

Find the volume of the solid obtained

- | | | |
|------|--|---|
| (b) | Five marbles are numbered 1, 2, 3, 4 and 5 and placed in bag. Two marbles are taken out in succession, the first marble not being replaced before the second is withdrawn. | |
| (i) | Find the probability that | |
| | (α) the 4 is selected. | 1 |
| | (β) the 3 is NOT selected. | 1 |
| | (γ) the 1 is the second marble chosen. | 1 |
| (ii) | If the first marble is replaced before the second is selected, find the probability that at least one 2 is drawn. | 2 |

Question 4 continued

(c) (i) Differentiate $y = \ln(\cos x)$ 2

(ii) Hence find $\int \tan x \, dx$ 1

(d) Find $\int_0^{\frac{\pi}{6}} \sec^2 2x \, dx$ 2

(e) For $a > 0$, it is given that 3

$$\int_1^4 \frac{4x}{4+x^2} \, dx = \ln a$$

By evaluating the integral, write down the value of a .

End of Section B

Section C (Use a SEPARATE writing booklet)

Question 5 (16 marks)

Marks

(a) Find the second derivative of $x \sin x$. 2

(b) Find $\int_0^1 \frac{e^x}{1+e^x} dx$ 2

(c) A continuous curve $y = f(x)$ has the following properties over the interval $a \leq x \leq b$:

$$f(x) > 0, f'(x) > 0, f''(x) > 0$$

(i) Sketch a curve satisfying these properties. 2

(ii) State the least value of $f(x)$ over this interval. 1

(d) At every point on a curve $\frac{d^2y}{dx^2} = 2x$. 4

The point $(3, 6)$ lies on the curve and its tangent at this point is inclined at 45° to the positive direction of the x axis.

Find the equation of the curve

(d) (i) For the function $f(x) = e^{x^2}$, copy and complete the following table, using 4 significant figures where necessary. 2

x	-1	-0.5	0	0.5	1
$f(x)$					

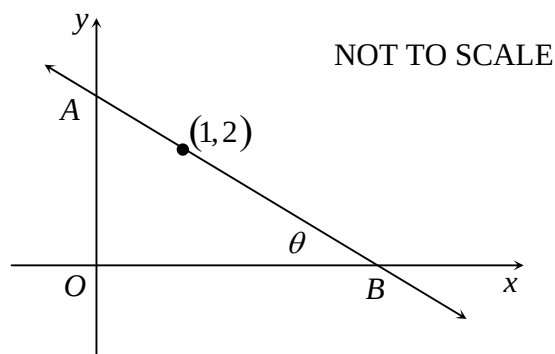
(ii) Hence, use Simpsons rule with 5 function values to approximate 3

$$\int_{-1}^1 e^{x^2} dx$$

Leave your answer correct to 3 significant figures.

Question 6 (19 marks)**Marks**

- (a) (i) Sketch $y = \log_e x$ and $y = x$ on the same diagram. 2
- (ii) Use your diagram in (i) to indicate the number of solutions to the equation $\log_e x = x$ 1
- (b) Given the curve $y = 2xe^{\frac{x}{2}}$
- (i) Show that $\frac{dy}{dx} = (x+2)e^{\frac{x}{2}}$ 2
- (ii) Find $\frac{d^2y}{dx^2}$ 2
- (iii) Find the minimum value of $2xe^{\frac{x}{2}}$ 2
- (iv) For what values of x is the curve concave down? 1
- (v) For what values of c does the equation $2xe^{\frac{x}{2}} = c$ have two unequal real roots? 1
- (c) In the diagram AB meets the x and y axes at B and A respectively and it passes through the point $(1, 2)$. The angle OBA is θ in radians.



- (i) Show that the equation of AB is given by 2
- $$y = -(\tan \theta)x + 2 + \tan \theta$$
- (ii) Find the area of $\triangle OAB$ in terms of $\tan \theta$. 2
- (iii) Find the value of θ , correct to 2 decimal places, for which the area is a minimum. 4

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

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SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2005
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #3

Mathematics

Sample Solutions

Section	Marker
A	AF
B	RB
C	RD

Question 1

$$(a) 150 \times \frac{\pi}{180} = \frac{5\pi}{6}^\circ$$

$$(b) \theta = 40 \times \frac{\pi}{180} = \frac{2\pi}{9}^\circ$$

$$\text{Area} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times (16)^2 \times \frac{2\pi}{9}$$

$$= \frac{256\pi}{9} \text{ cm}^2$$

$$\approx 89 \text{ cm}^2$$

$$(c)(i) \frac{1}{\sqrt{e}} \approx 0.61$$

$$(ii) \cos 3 = -0.99$$

$$(d)(i) -4e^{-4x}$$

$$(ii) \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$(e)(i) P(\text{no 4's}) = \frac{27}{30} \times \frac{26}{29}$$

$$= \frac{117}{145}$$

$$P(\text{at least one 4}) = 1 - \frac{117}{145}$$

$$= \frac{28}{145}$$

$$(f) f'(x) = 2x + 3$$

$$f(x) = x^2 + 3x + C$$

$$\text{since } f(1) = 6$$

$$6 = (1)^2 + 3(1) + C$$

$$C = 2$$

$$f(x) = x^2 + 3x + 2$$

$$f(2) = (2)^2 + 3(2) + 2 \\ = 12$$

$$(g)(i) \angle BED = 150^\circ$$

(co-interior angles are supplementary)
 $BE \parallel CD$

$$(ii) \text{ In } \Delta\text{'s } BEA \text{ \& } CDA$$

$\angle CAE$ is common

$$\angle BEA = \angle CDA \text{ (corresponding angles)} \\ BE \parallel CD$$

$$\therefore \Delta BEA \sim \Delta CDA$$

$\therefore$ corresponding sides are in the same ratio

$$\frac{BE}{CD} = \frac{AB}{AC}$$

$$\frac{BE}{9} = \frac{2}{6}$$

$$BE = 3 \text{ cm}$$

Question 2

(a)(i) $y' = uv' + vu'$ $u = 2x$ $v = \ln x$
 $u' = 2$ $v' = \frac{1}{x}$

$$y' = 2x \cdot \frac{1}{x} + 2 \cdot \ln x$$

$$= 2 + 2 \ln x$$

(ii) $y' = \frac{vu' - uv'}{v^2}$ $u = \cos x$ $v = x$
 $u' = -\sin x$ $v' = 1$

$$y' = \frac{x(-\sin x) - 1 \cdot \cos x}{x^2}$$

$$= \frac{-x \sin x - \cos x}{x^2}$$

(b)(i) $\int_1^4 x^{-3/2} dx$

$$= \left[-2x^{-1/2} \right]_1^4$$

$$= -2(4)^{-1/2} - (-2(1)^{-1/2})$$

$$= -1 + 2$$

$$= 1$$

(ii) $\int \frac{x^3 + x}{x^2} dx$

$$= \int \left(x + \frac{1}{x} \right) dx$$

$$= \frac{x^2}{2} + \ln x + C$$

(iii) $\int (e^{2x} + 2e^x + 1) dx$

$$= \frac{1}{2} e^{2x} + 2e^x + x + C$$

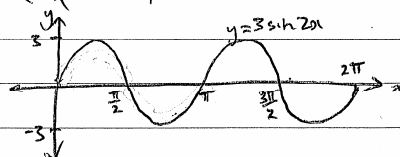
(c)(i) $\frac{dp}{dt} < 0$ from 1998 to mid 2002

$$\frac{dp}{dt} = 0 \text{ in mid 2002}$$

$$\frac{dp}{dt} > 0 \text{ from mid 2002 to now}$$

(ii) $\frac{d^2p}{dt^2} > 0$

(d)(i) amplitude = 3 period = $\frac{2\pi}{2} = \pi$



(ii) $\int_0^{\pi/4} 3 \sin 2x dx$

$$= \left[-\frac{3}{2} \cos 2x \right]_0^{\pi/4}$$

$$= -\frac{3}{2} \cos 2\left(\frac{\pi}{4}\right) - \left(-\frac{3}{2} \cos 0\right)$$

$$= -\frac{3}{2} \cos \frac{\pi}{2} + \frac{3}{2} \cos 0$$

$$= \frac{3}{2}$$

3) (a)(i) $y = x - 4$
 $y = -x^2 + 5x - 4$

let $x - 4 = -x^2 + 5x - 4$

$$x^2 - 4x = 0$$

$$x(x - 4) = 0$$

$$x = 0 \quad y = -4$$

$$A(0, -4)$$

$$x = 4, \quad y = 0$$

$$B(4, 0)$$

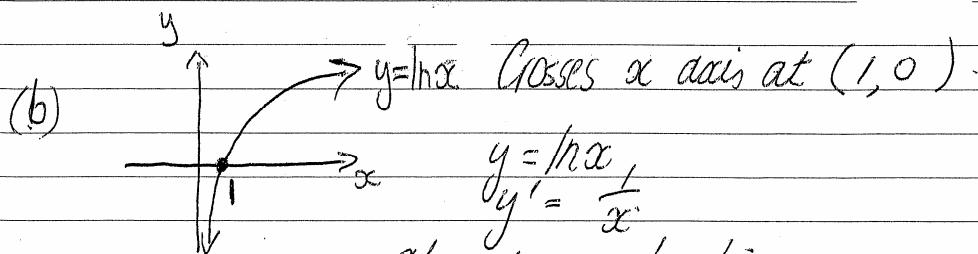
4

(ii) $A = \int_0^4 [(-x^2 + 5x - 4) - (x - 4)] dx$

$$= \int_0^4 (-x^2 + 4x) dx$$

$$= \left[-\frac{x^3}{3} + 2x^2 \right]_0^4 = -\frac{4^3}{3} + 2 \times 4^2 - 0$$

$$= -\frac{64}{3} + 32 = 10\frac{2}{3} \text{ u}^2 //$$



$$y = \ln x$$

$$y' = \frac{1}{x}$$

$$\text{At } x = 1, m = \frac{1}{1} = 1$$

Using $(y - y_1) = m(x - x_1)$

$$y - 0 = 1(x - 1)$$

$$y = x - 1 //$$

$$\text{or } x - y - 1 = 0$$

$$(c) \quad y = x^3 - 6x^2 + 9x + 6$$

$$y' = 3x^2 - 12x + 9$$

$$y'' = 6x - 12$$

(i) Stat pts exist when $y' = 0$ $3x^2 - 12x + 9 = 0$

$$3(x^2 - 4x + 3) = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x=3 \quad y = 27 - 64 + 27 + 6 = 6 \quad (3, 6)$$

$$x=1 \quad y = 1 - 6 + 9 + 6 = 10 \quad (1, 10)$$

At $(3, 6)$ $y'' = 6 \times 3 - 12 > 0$ min stat pt.

At $(1, 10)$ $y'' = 6 \times 1 - 12 < 0$ max stat pt.

(ii) inflexions occur when $y'' = 0$ and there is a sign change.

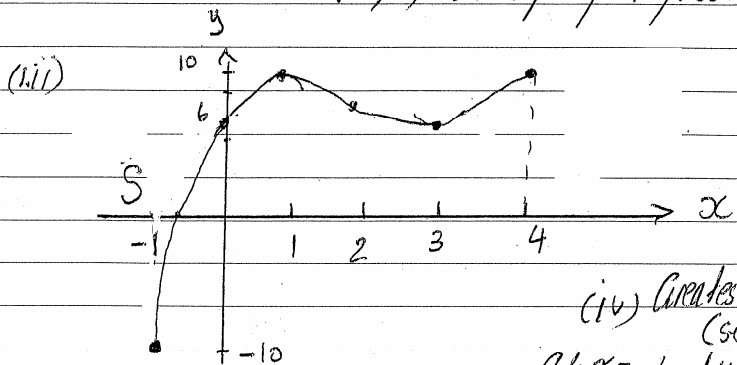
$$6x - 12 = 0$$

$$6(x-2) = 0$$

$$x=2, \quad y = 8 - 6 \times 4 + 18 + 6 = 8 \quad (2, 8)$$

At $(2, 8)$ $x=2-\epsilon \quad y'' < 0$ } sign change
 $x=2+\epsilon \quad y'' > 0$ }

$(2, 8)$ is a pt of inflexion

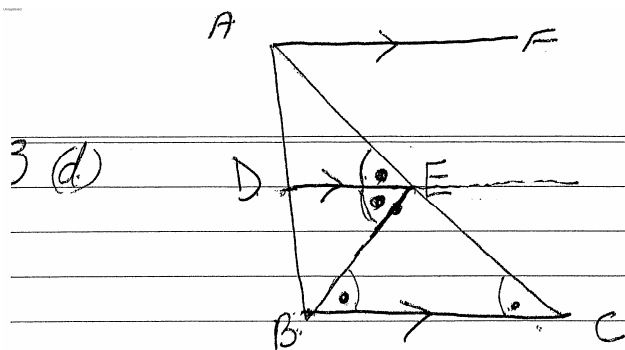


$$\text{At } x=-1, y = -1 - 6 - 9 + 6 = -10$$

$$\text{At } x=4, y = 64 - 96 + 36 + 6 = 10$$

(iv) Greatest rate of increase is
 (see diagram)

At $x=-1$ $|y'|$ is greatest.



(i) $\angle EBC = \angle BED$ alternate angles $DE \parallel BC$
 $\angle BCE = \angle AED$ corresponding angles $DE \parallel BC$
 $\therefore \triangle BEC$ is isosceles, $\angle EBC = \angle BCE$

(ii) Why does $\frac{AD}{DB} = \frac{AE}{EC}$ ratio of intercepts result,
 as $AF \parallel DE \parallel BC$ and
 AC is the transversal

(iii) Since $BE = EC$ from (i)
 $\frac{AD}{DB} = \frac{AE}{EC}$ from (ii)

So $\frac{AD}{DB} = \frac{AE}{EB}$

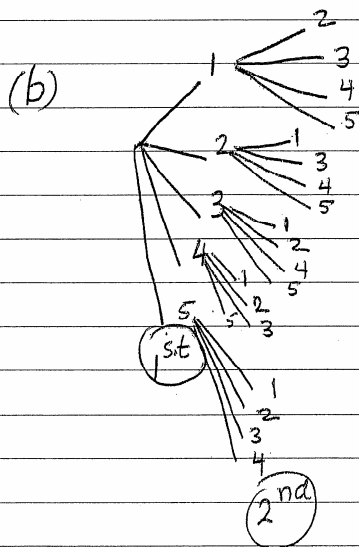
$$(4) (a) V = \pi \int_0^1 \left(\frac{1}{\sqrt{2x+1}} \right)^2 dx$$

$$= \pi \int_0^1 \frac{1}{2x+1} dx$$

$$= \frac{\pi}{2} \ln(2x+1) \Big|_0^1$$

$$= \frac{\pi}{2} [\ln(3) - \ln(1)]$$

$$= \frac{\pi}{2} \ln 3 \approx 1.73$$



$$(i) (a) \frac{12}{20} = \frac{3}{5}$$

$$(b) \frac{12}{20} = \frac{3}{5}$$

$$(c) \frac{4}{20} = \frac{1}{5}$$

(ii) $P(\text{at least one 2}) = \text{one 2 on either marble or both 2}$

$$\frac{9}{25}$$

$$(c) \quad (i) \quad y = \ln(\cos x)$$

$$y' = \frac{1}{\cos x} \times -\sin x = \frac{-\sin x}{\cos x} = -\tan x$$

$$(ii) \quad -\int -\tan x \, dx = -\ln(\cos x) + C$$

(d)

$$\frac{1}{2} \int_0^{\frac{\pi}{6}} 2 \sec^2 2x \, dx = \left[\frac{1}{2} \tan 2x \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{2} (\tan \frac{\pi}{3} - \tan 0)$$

$$= \frac{1}{2} (\sqrt{3} - 0)$$

$$= \frac{\sqrt{3}}{2} \text{ or } \approx 0.866$$

$$(e) \quad \int_1^4 \frac{4x}{4+x^2} \, dx$$

$$\Rightarrow 2 [\ln 20 - \ln 5]$$

$$\frac{4}{2} \int_1^4 \frac{2x}{4+x^2} \, dx$$

$$= 2 \ln \left(\frac{20}{5} \right)$$

$$= 2 \ln 4$$

$$= 2 \ln(4+x^2) \Big|_1^4$$

$$= \ln 4^2 = \ln 16$$

$$= 2 \ln 20 - 2 \ln 5$$

$$\text{So } a = 16 //$$

Q5(a) $y = x \sin x$

$$y' = \sin x \cdot 1 + x \cdot \cos x$$

$$= \sin x + x \cos x$$

$$y'' = \cos x + (\cos x \cdot 1 + x \cdot -\sin x)$$

$$= 2 \cos x - x \sin x$$

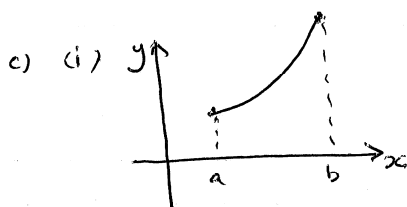
(b) $\int_0^1 \frac{e^x}{1+e^x} dx$

$$= [\ln(1+e^x)]_0^1$$

$$= \ln(1+e) - \ln(1+1)$$

$$= \ln(1+e) - \ln 2$$

$$= \ln\left(\frac{1+e}{2}\right)$$



(ii) $f(a)$

(d) $y'' = 2x$

$$y' = x^2 + C$$

When $x=3$: $1 = 9 + C$

$$\therefore C = -8$$

$$\therefore y' = x^2 - 8$$

$$y = \frac{1}{3}x^3 - 8x + C_1$$

When $x=3$: $6 = 9 - 24 + C_1$

$$C_1 = 21$$

$$\therefore y = \frac{1}{3}x^3 - 8x + 21$$

(d) $f(x) = e^{x^2}$

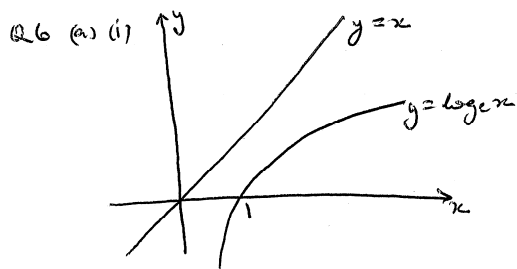
x	-1	-0.5	0	0.5	1
$f(x)$	2.718	1.284	1	1.284	2.718

(ii) $\int_{-1}^1 e^{x^2} dx \approx \frac{0.5}{3} (2.718 + 4 \times 1.284$

$$+ 2 \times 1 + 4 \times 1.284 + 2.718)$$

$$= 2.9513333 \dots$$

$$\approx 2.95$$



(ii) 0

(b) (i)

$$y = 2x \cdot e^{x/2}$$

$$y' = e^{x/2} \cdot 2 + 2x \cdot e^{x/2} \cdot \frac{1}{2}$$

$$= e^{x/2} \cdot 2 + x \cdot e^{x/2}$$

$$= (x+2)e^{x/2}$$

(ii)

$$y'' = e^{x/2} \cdot 1 + (x+2) \cdot e^{x/2} \cdot \frac{1}{2}$$

$$= e^{x/2} + \frac{1}{2}x e^{x/2} + e^{x/2}$$

$$= e^{x/2} \left(\frac{1}{2}x + 2 \right)$$

(iii) For min value, consider $y' = 0$

$$\therefore x = -2$$

When $x = -2$ $y'' = e^{-1}(-1+2)$

$$= e^{-1} > 0$$

$\therefore$ Min value when $x = -2$

$\therefore$ Min value is $2x - 2x e^{-1}$

$$= -4(e^{-1}) \quad (\approx -1.4715)$$

(iv) Concave down $\Rightarrow y'' < 0$

$$\therefore \frac{1}{2}x + 2 < 0$$

$$\therefore \frac{1}{2}x < -2$$

$$\therefore x < -4$$

(v) 2 unequal roots if $-4e^{-1} < c < 0$

(c) (i) $m = \tan(\pi - \theta)$

$$= -\tan \theta$$

$\therefore$ Eqn is $y - 2 = -\tan \theta (x - 1)$

$$\therefore y - 2 = -\tan \theta x + \tan \theta$$

$$\therefore y = -\tan \theta x + 2 + \tan \theta$$

(ii) If $x = 0$, $y = 2 + \tan \theta \Rightarrow A(0, 2 + \tan \theta)$

If $y = 0$, $\tan \theta x = 2 + \tan \theta$

$$x = \frac{2 + \tan \theta}{\tan \theta} \Rightarrow B\left(\frac{2 + \tan \theta}{\tan \theta}, 0\right)$$

$\therefore$ Area of $\triangle AOB = \frac{1}{2} \times \frac{2 + \tan \theta}{\tan \theta} (2 + \tan \theta)$

$$= \frac{(2 + \tan \theta)^2}{2 \tan \theta}$$

(iii) $\frac{dA}{d\theta} = \frac{(2 \tan \theta \cdot 2(2 + \tan \theta) \sec^2 \theta - (2 + \tan \theta)^2 \cdot 2 \sec^2 \theta)}{4 \tan^2 \theta}$

$$= \frac{2 \sec^2 \theta (2 + \tan \theta)(2 \tan \theta - (2 + \tan \theta))}{4 \tan^2 \theta}$$

$$= \frac{2 \sec^2 \theta (2 + \tan \theta)(\tan \theta - 2)}{4 \tan^2 \theta}$$

$$= \frac{(2 + \tan \theta)(\tan \theta - 2)}{2 \sin^2 \theta}$$

Consider $\frac{dA}{d\theta} = 0$

$$\therefore \tan \theta = 2 \text{ or } \tan \theta = -2$$

$$\therefore \tan \theta = 2 \quad (\text{as } 0 < \theta < \frac{\pi}{2})$$

$$\therefore \theta = 1.107148 \dots$$

θ	1	1.107	2
$\frac{dA}{d\theta}$	-ve	0	+ve
Curve	\	-	/

$\therefore$ Min value occurs when $\theta \approx 1.1$